Manufacturing Downtime & Yield Analytics

Manufacturing BI Engineering Project

TABLEAU • MICROSOFT POWER BI • QLIK SENSE

Hemanth V
Naina Karn
Nikhat Ansar
Vrinda Rajput
Abhinav Yadav

A Larsen & Toubro Group Company

LTM

Project overview

Manufacturing downtime

Time when a machine is stopped, unavailable or not producing as planned.

Why monitoring matters

Shows production efficiency, scrap loss, downtime impact and assets needing attention.

Production yield

Percentage of total production that becomes acceptable good output.

Problem addressed

Raw records contained duplicates, inconsistent values, missing references and injected quality issues.

Project outcome

Cleaned data, shared dimensional model and interactive analysis in Tableau, Power BI and Qlik Sense.

Production + Quality + Downtime → trusted operational decisions

Project objectives

1 **Production**

Analyse output and trends

2 **Quality**

Calculate yield and scrap %

3 **Downtime**

Measure events, hours and reasons

4 **Comparison**

Compare plant, line and machine

5 **Root cause**

Identify major contributors

6 **Dashboards**

Build in three BI platforms

7 **Decisions**

Support operational reviews

Dataset overview

Dataset	Purpose	Documented raw	Verified analytical view
production_batches	Batch output, good and scrap quantities	10,200	10,000
machine_downtime	Machine stoppage events and reasons	18,180	18,000
machines	Machine master and type	250	251
plants	Plant master and location	25	26
shift_calendar	Shift and operational schedule	720	721
calendar	Date analysis layer	120	122
data_quality_injections	Injected quality test records	5	Removed / not retained

Cleaned dimensions include UNKNOWN rows where required. Injected test records were removed where appropriate.

ETL process and data preparation



BI-native workflow only

- Power Query for Power BI transformations
- Tableau Prep for Tableau preparation
- Qlik Load Script for Qlik Sense

Data cleaning and validation

Duplicate removal

- Latest batch_id / downtime_id by created_ts
- Retain latest complete record
- Preserve flags for traceability

Standardisation & missing values

- Normalise dates and quantities
- Map DL, GJ and BR to full state names
- Use UNKNOWN members instead of deleting

Production validation

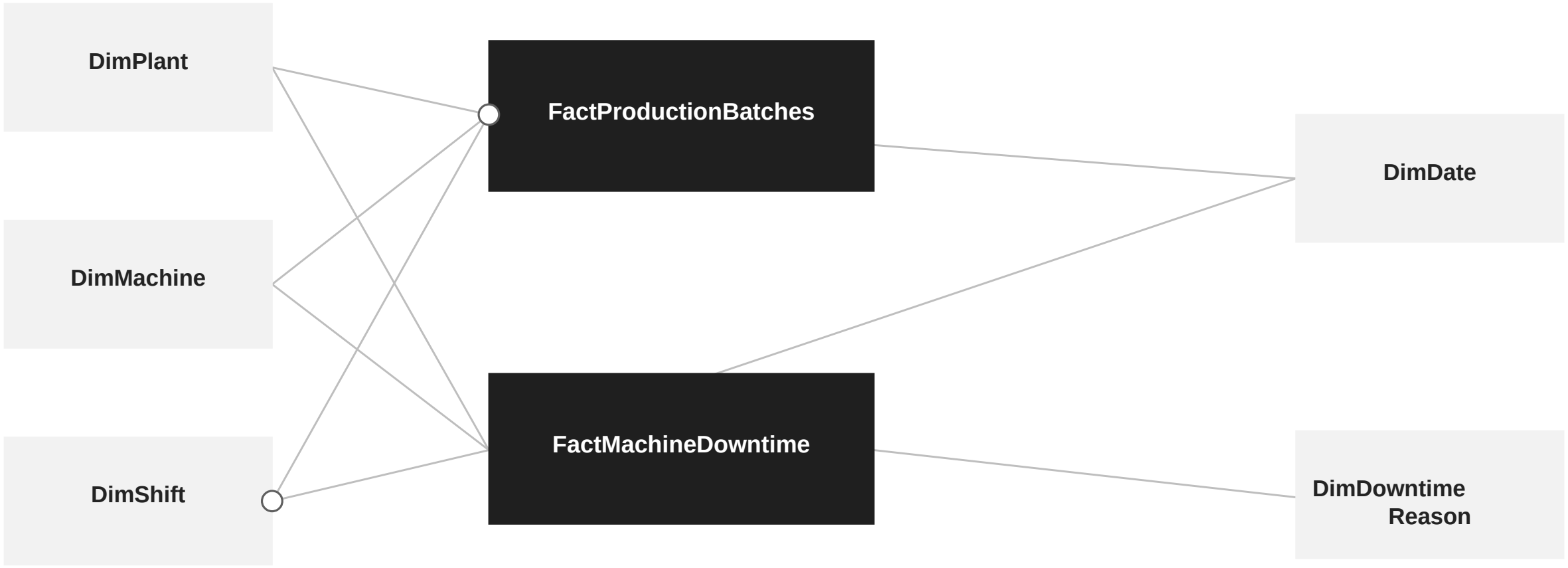
- Check missing or negative quantities
- Check scrap ≤ total
- Reconcile total = good + scrap
- Assign PASS or REVIEW

Downtime validation

- Check start/end sequence
- Derive duration where possible
- Compare source vs derived duration
- Flag unusual values for review

Outcome: auditable records, explicit exceptions and safer KPI aggregation

Data model: Galaxy Schema



Shared: Plant, Machine, Shift and Date | Downtime Reason: downtime fact only

Two fact tables remain separate because production batches and downtime events have different grains.

Internal, Client Confidential

KPIs and analytical approach

7.29M	7.01M	287.85K	96.05%	10.95K
Total Production	Good Quantity	Scrap Quantity	Overall Yield	Downtime Hours

Core formulas

- Yield % = Good Quantity ÷ Total Production × 100
- Scrap % = Scrap Quantity ÷ Total Production × 100
- Downtime Hours = Valid Downtime Minutes ÷ 60
- Average Downtime = Valid Downtime Minutes ÷ Valid Events
- Uptime % = Available Time less Downtime ÷ Available Time × 100

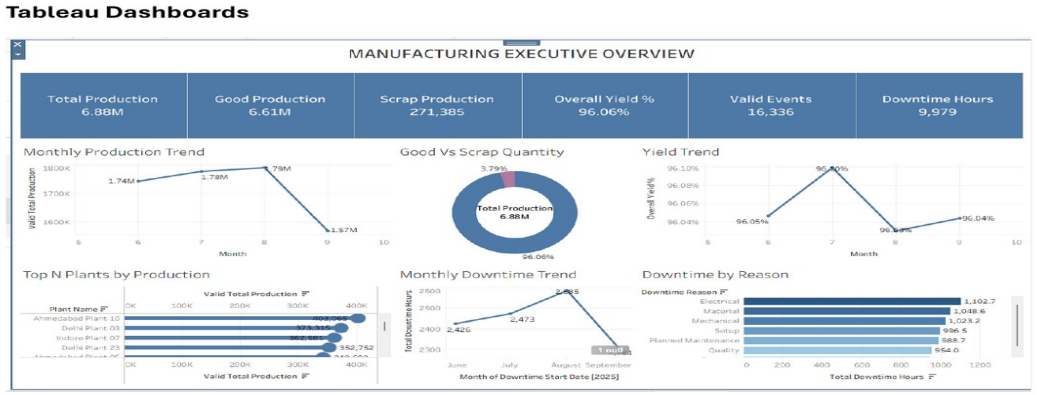
Additional analysis

- Total downtime events
- Plant yield ranking
- Planned vs unplanned downtime
- Machine and shift comparisons
- PASS records for trusted production KPIs

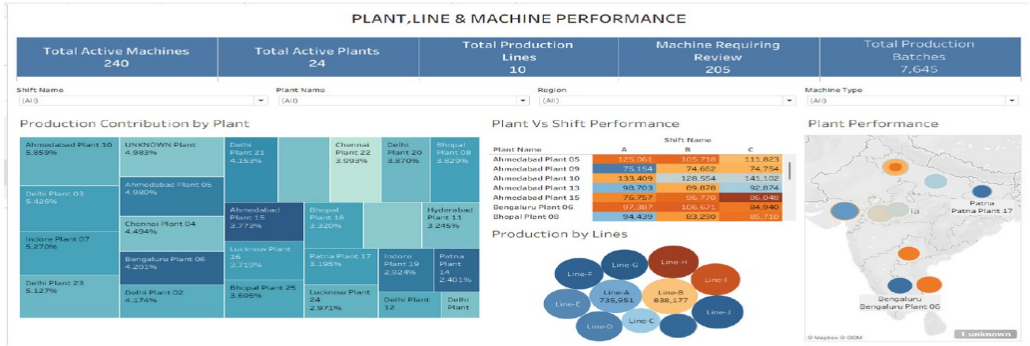
Validate logic across all three BI platforms

Tableau dashboards

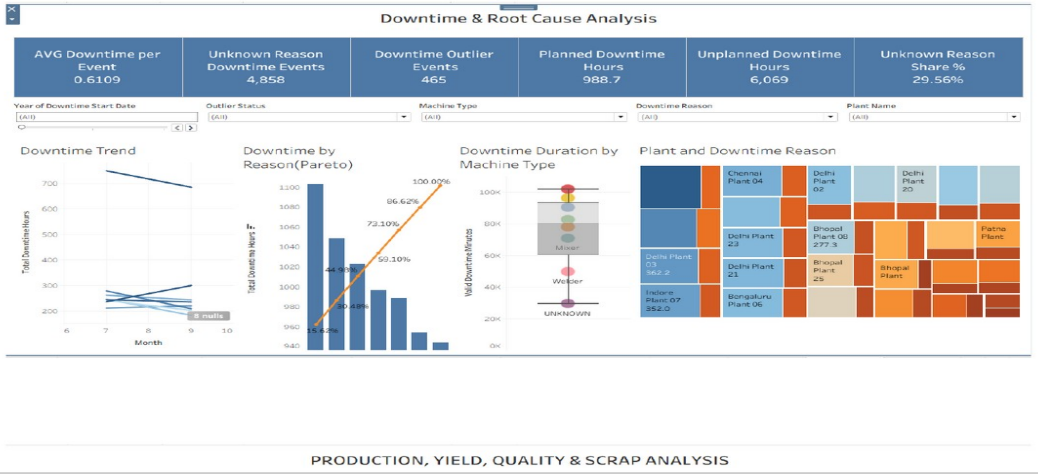
Manufacturing Executive Overview



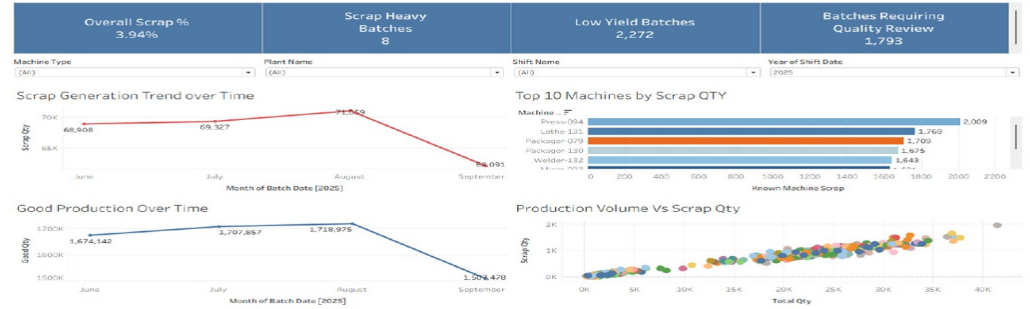
Plant, Line & Machine Performance



Downtime & Root Cause Analysis



Production Yield & Scrap Analysis

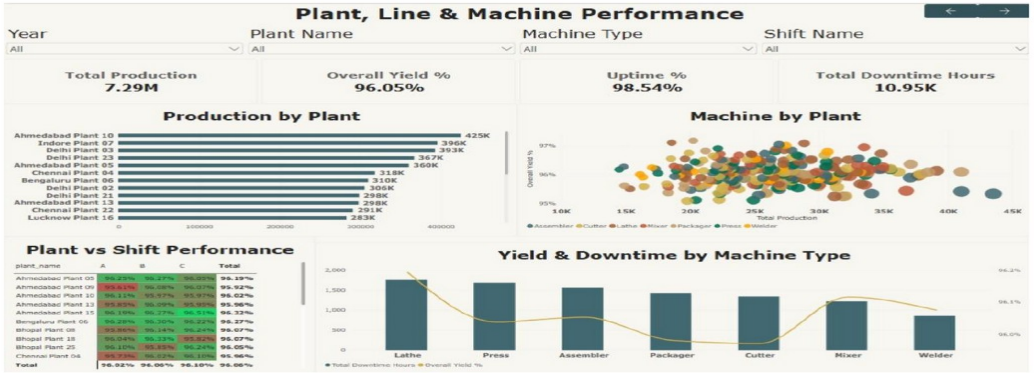


Microsoft Power BI dashboards

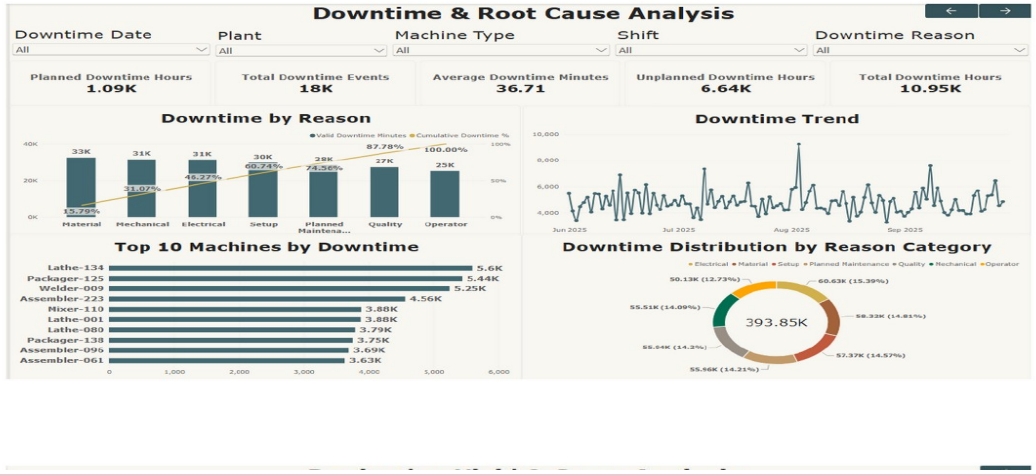
Manufacturing Executive Overview



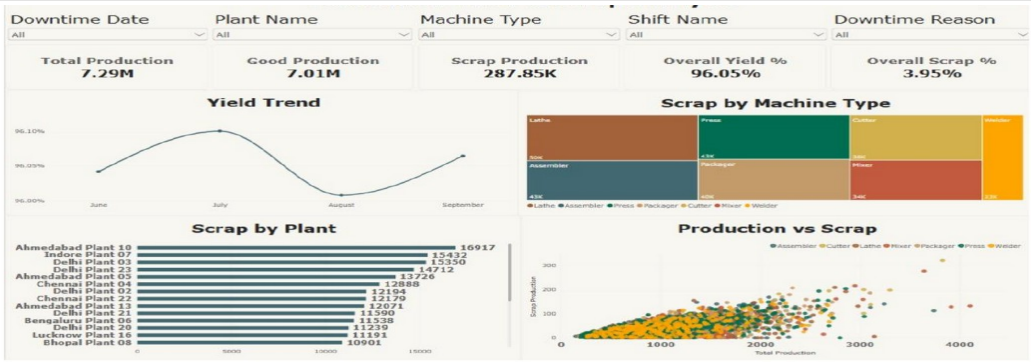
Plant, Line & Machine Performance



Downtime & Root Cause Analysis

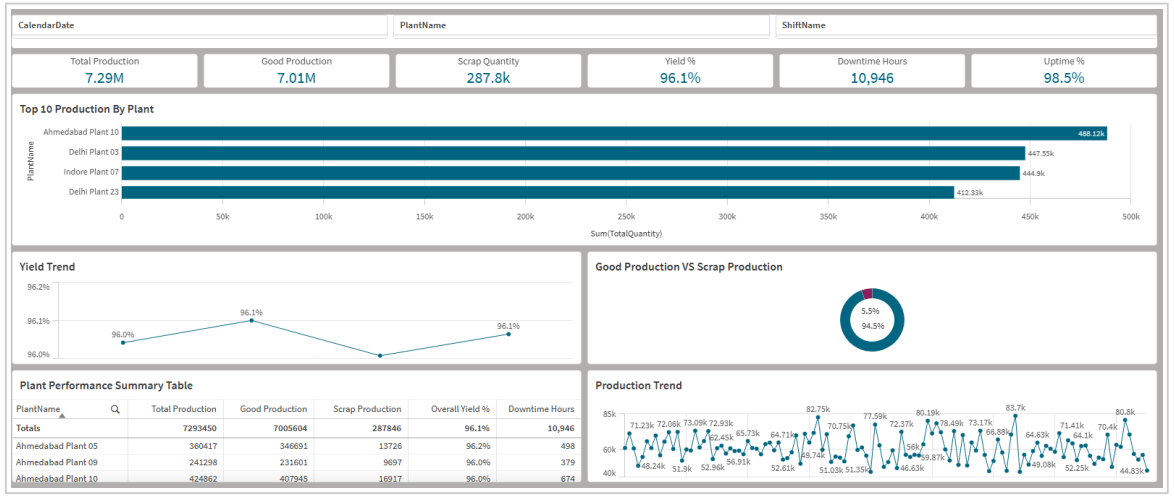


Production Yield & Scrap Analysis

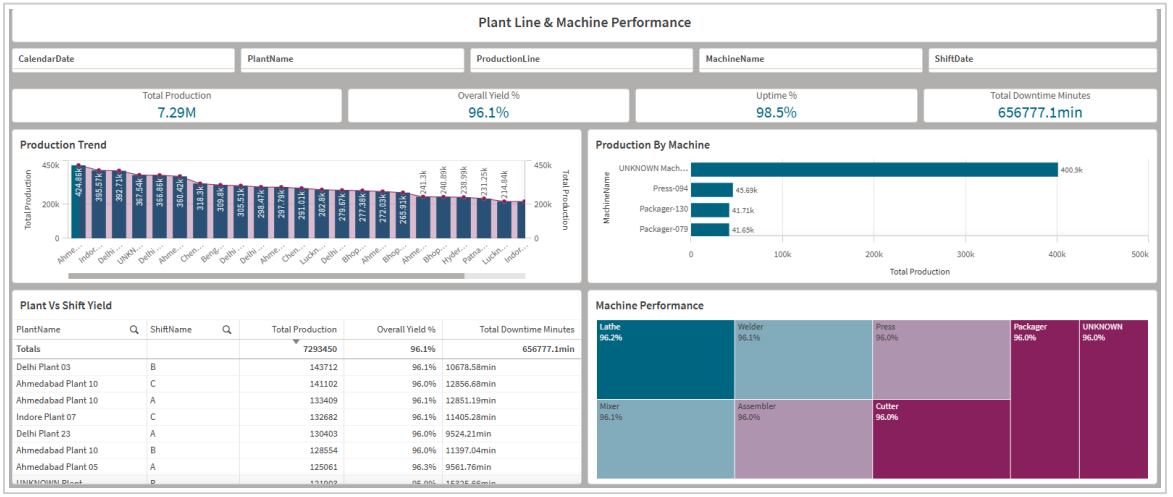


Qlik Sense dashboards

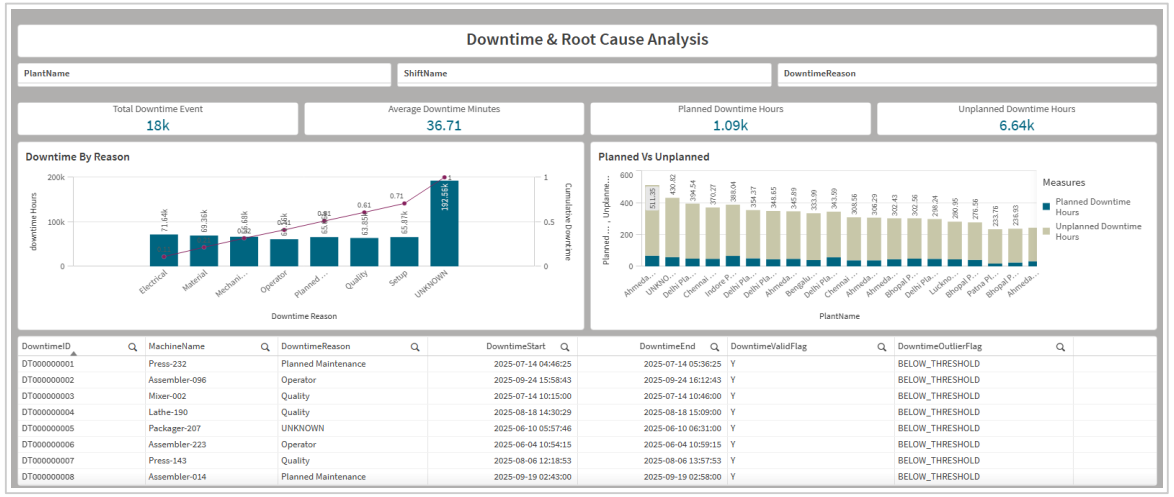
Manufacturing Executive Overview



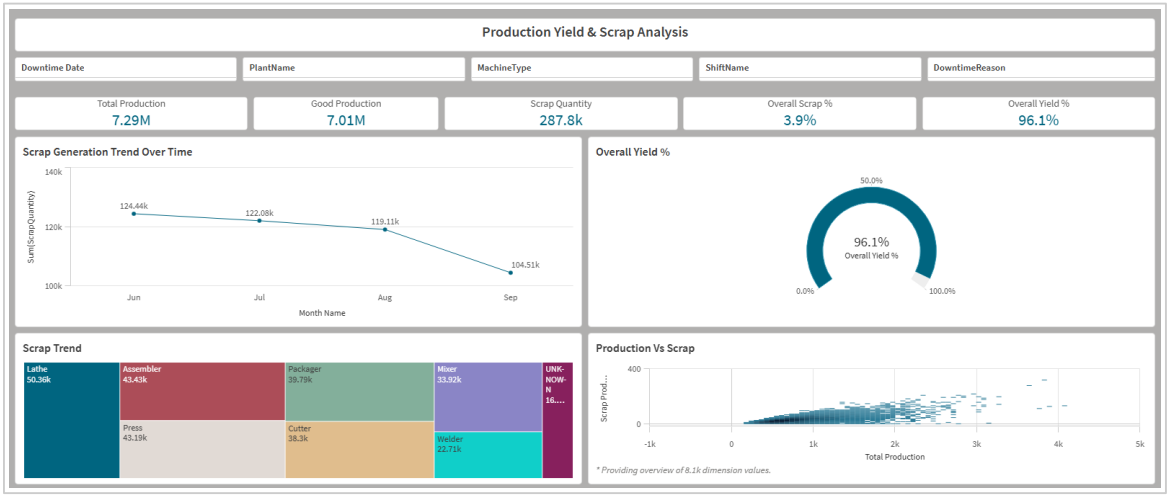
Plant, Line & Machine Performance



Downtime & Root Cause Analysis



Production Yield & Scrap Analysis



Key insights and findings

7.29M	96.05%	287.85K	10.95K hrs	98.54%
Total production	Overall yield	Scrap	Downtime	Uptime

Performance

Good output is 7.01M; scrap represents 3.95% of production.

Downtime

18K events average 36.71 minutes; unplanned hours (6.64K) materially exceed planned hours (1.09K).

Priority assets

Lathe-134 leads displayed machine downtime at 5.6K minutes; Ahmedabad Plant 10 leads production at about 425K.

Data quality

Tableau PASS-filtered scope shows 6.88M production, 16,336 valid events and 29.56% unknown-reason share.

Business recommendations

1 Prioritise chronic downtime

Review Lathe-134 and other top machines; validate maintenance history and recurring reasons.

2 Focus on unplanned loss

Use Pareto views to target Electrical, Material, Mechanical and Setup contributors.

3 Improve reason-code capture

Reduce UNKNOWN reasons with mandatory codes, controlled lists and supervisor checks.

4 Control scrap-heavy batches

Track high-scrap machines and batches; compare shift, line and product context.

5 Operationalise reviews

Use weekly plant and machine scorecards with PASS/REVIEW data-quality visibility.

6 Keep cross-tool reconciliation

Maintain one KPI definition sheet and test totals after every refresh.

Limitations and future scope

Current limitations

- Analysis is limited to the provided files
- No real-time machine monitoring
- Some records require data-quality review
- Uptime uses a simplified available-time calculation
- Predictive maintenance is not implemented

Future scope

- Automated and scheduled data refresh
- Live manufacturing data integration
- Better downtime reason-code completeness
- Alerts for high downtime or scrap
- Predictive maintenance after sufficient history
- More robust OEE with planned production time

Conclusion

- Converted raw manufacturing data into cleaned and validated analytical data
- Built a Galaxy Schema with separate production and downtime facts
- Analysed yield, scrap, machine downtime and root causes
- Delivered four dashboards each in Tableau, Power BI and Qlik Sense
- Created trusted visibility to support better operational decisions

Thank You

Hemanth V • Naina Karn • Nikhat Ansar
Vrinda Rajput • Abhinav Yadav